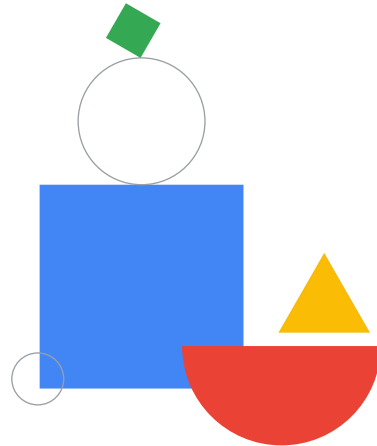


Preparing for Your Associate Cloud Engineer Journey

Section 5: Configuring Access and Security



Welcome to section 5 of the exam guide: Configuring Access and Security.



Module agenda

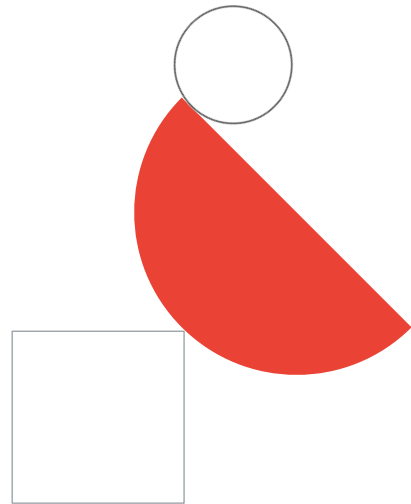
- 01 Managing access for Cymbal Superstore's cloud solutions
- 02 Diagnostic questions
- 03 Review and study planning

In this module you'll explore the scope of configuring access and security. This involves managing IAM as well as service accounts for cloud solutions. These areas correspond to the fifth (and last) section of the Associate Cloud Engineer Exam Guide.

We'll start by discussing your role as an Associate Cloud Engineer in managing access for Cymbal Superstore's cloud solutions. Next, you'll assess your skills in this section of the exam guide through 5 diagnostic questions. There are fewer questions for this module. While this is an important area for you to understand as an Associate Cloud Engineer, it is narrower in scope.

When we review the questions, identify the resources you'll want to include in your study plan.

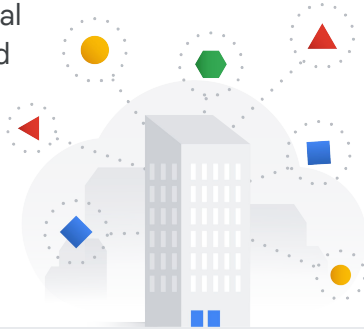
Managing access for Cymbal Superstore's cloud solutions



As Cymbal Superstore uses its applications on Google Cloud, an Associate Cloud Engineer plays an ongoing role in configuring and managing IAM access and service accounts. Let's explore some examples of how you might do this at Cymbal Superstore.

The next step:

ongoing access and security for Cymbal Superstore's cloud solutions



- Managing Identity and Access Management (IAM)
- Managing service accounts



To successfully perform the Associate Cloud Engineer role at Cymbal Superstore, you need to be able to manage Identity and Access Management (IAM) in Google Cloud. We talked about the basics of IAM in the first module from the perspective of setting up cloud projects and accounts. Here, you'll consider skills involved in managing access. You'll also need to be familiar with service accounts and recommended practices to manage them in Google Cloud.

Setting up a service account for Cymbal Superstore's supply chain app



- 1 Create a service account
- 2 Assign Permissions
- 3 Attach to a VM

To give you a better idea of what configuring access and security involves in practice, let's explore an example of where you might use a service account at Cymbal Superstore.

Cymbal Superstore's supply chain app is built on a LAMP stack using Compute Engine VM instances. It uses Cloud SQL as a backing data store. The app needs to talk to Cloud SQL to update inventory levels. It does this through a service account attached to the VM it runs on.

Service accounts are designed to enable machine-to-machine communication for just this purpose.

The first step in setting up a service account for Cymbal Superstore's supply chain app is to create the service account.

Next, you assign permissions to the service account you just created.

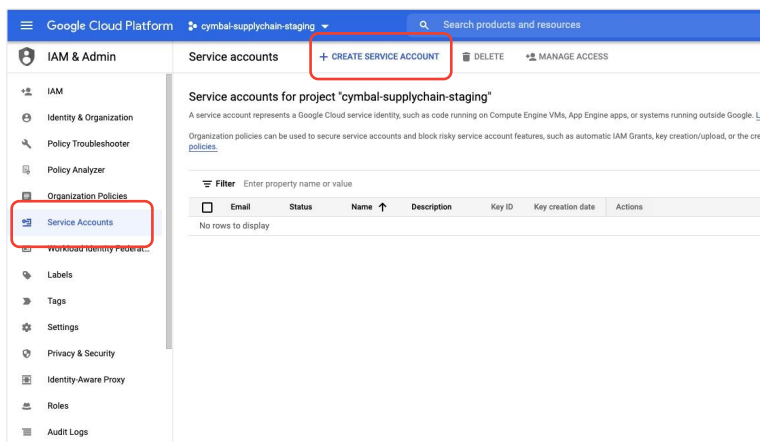
Finally, you attach that service account to a Compute Engine VM. Attaching a service account allows the VM and all the apps running on it use the permissions assigned to the service account.

Let's look at these steps in more detail.

01

Create a service account:

Where to look



Go to the project you want to add the service account to. Service accounts are both identities and managed resources in Google Cloud. Select the **service account** link in the IAM menu of your project.

Then select **create service account**.

01

Create a service account: Enter service account details

Google Cloud Platform cymbal-supplychain-staging Search products and resources

IAM & Admin

Create service account

1 Service account details

Service account name
vm-service-account

Display name for this service account

Service account ID
vm-service-account @helpful-chiller-328713.iam.gserviceaccount.com X ↺

Service account description
service account to be attached to vm's for supply chain app

Describe what this service account will do

CREATE AND CONTINUE

2 Grant this service account access to project (optional)

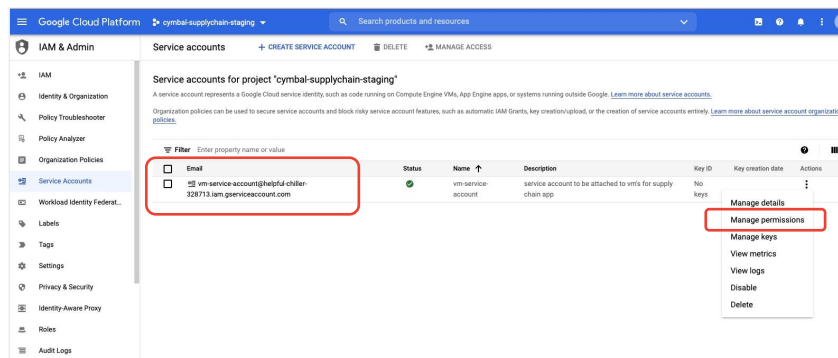
3 Grant users access to this service account (optional)

DONE CANCEL

In the dialogue that comes up, name your service account and note the email address associated with it. You can also provide a description of what this service account does.

02

Assign permissions: Where to look



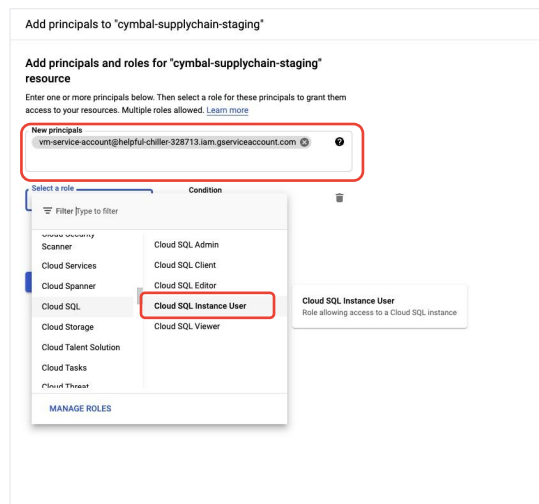
Once you select **create**, your new service account will be added to the list of all of your service accounts..

Select the three ellipses under actions and for a list of all the actions you can perform on your new service account.

Next, we'll use one of these choices to manage permissions for the service account.

02

Assign permissions: Add necessary permissions



Select **manage permissions** under the actions dialogue in the service account list. A new menu lets you pick your service account and add permissions to it.

Copy your service account email address identifier. Search or browse the permissions to find the ones you need to add. In our example, we'll give our service account permissions as a Cloud SQL instance User.

03

Add to a VM instance

Where to look

Identity and API access ⓘ

Service accounts ⓘ

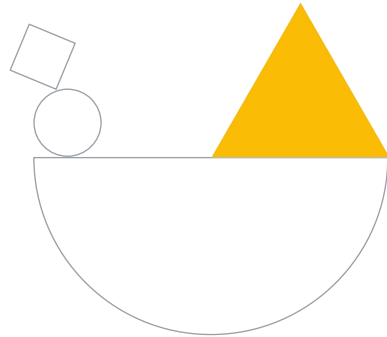
Service account
vm-service-account ▼

Access scopes ⓘ

Use IAM roles with service accounts to control VM access. [Learn more](#)

Finally, when you add your VM instance you have a chance to add the service account to it under the Identity and API access section. This covers authorization. Authentication is another important aspect of both user accounts and service accounts that you should be familiar with as an Associate Cloud Engineer.

Diagnostic questions



Now that you have some context for the objectives in this exam section, it's time to take a self-assessment focused on configuring access and security for Google Cloud solutions.

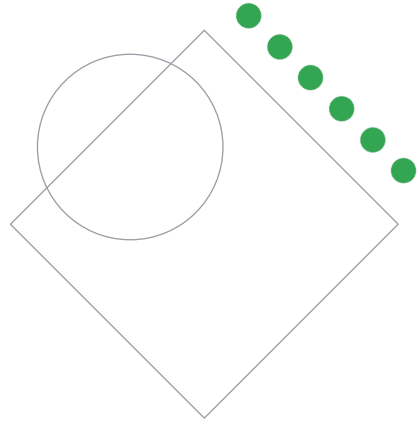
Please complete the diagnostic questions now

- The diagnostic questions are available in the workbook.



Please take 15 minutes to complete the diagnostic questions for this section.

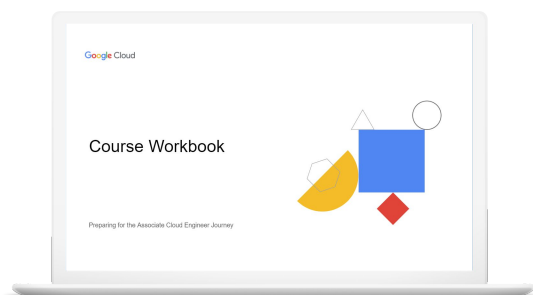
Review and study planning



What areas do you need to develop your skills in order to successfully manage access for Google Cloud solutions? Recall that this exam section is narrower in scope but nevertheless important for the role of an Associate Cloud Engineer. Let's review the diagnostic questions to help you target your study time to focus on the areas where you need to develop your skills.

Your study plan:

Configuring access and security



5.1

Managing Identity and Access Management (IAM)

5.2

Managing service accounts

We'll approach this review by looking at the objectives of this exam section and the questions you just answered about each one. We'll introduce an objective, briefly review the answers to the related questions, then talk about where you can find out more in the learning resources and/or in Google Cloud documentation. As we go through each section objective, use the page in your workbook to mark the specific documentation, courses (and modules!), and skill badges you'll want to emphasize in your study plan.

There are fewer objectives and tasks involved with this section - but don't forget to plan for them in your study time.

5.1 | Managing Identity and Access Management (IAM)

Considerations include:

- Viewing and creating IAM policies
- Managing the various role types and defining custom IAM roles (e.g., basic, predefined and custom)

We worked on planning Cymbal Superstore's resource hierarchy earlier in the course. The organization is divided into folders, and the folders into projects. Identity and Access Management lets your users and groups access Google Cloud resources. As an Associate Cloud Engineer, it is important that you know both what users and groups to implement for Cymbal Superstore's needs and how to implement them.

The tasks included in this part of your job as an Associate Cloud Engineer include viewing IAM policies, creating IAM policies, and knowing when to implement the different types of policies, including basic, predefined and custom roles.

These are the diagnostic questions you answered that relate to this area:

Question 1: Identify types of members you can assign access to in IAM.

Question 2: Describe how to assign roles in the IAM interface.

Question 3: List the steps to create a custom role in IAM.

5.1 | Diagnostic Question 01 Discussion



You need to configure access to Spanner from the GKE cluster that is supporting Cymbal Superstore's ecommerce microservices application. You want to specify an account type to set the proper permissions.

What should you do?

- A. Assign permissions to a Google account referenced by the application.
- B. Assign permissions through a Google Workspace account referenced by the application.
- C. Assign permissions through service account referenced by the application.
- D. Assign permissions through a Cloud Identity account referenced by the application.

Question:

You need to configure access to Spanner from the GKE cluster that is supporting Cymbal Superstore's ecommerce microservices application. You want to specify an account type to set the proper permissions. What should you do?

5.1 Diagnostic Question 01 Discussion



You need to configure access to Spanner from the GKE cluster that is supporting Cymbal Superstore's ecommerce microservices application. You want to specify an account type to set the proper permissions.

What should you do?

- A. Assign permissions to a Google account referenced by the application.
- B. Assign permissions through a Google Workspace account referenced by the application.
- C. Assign permissions through service account referenced by the application. ✓
- D. Assign permissions through a Cloud Identity account referenced by the application.

Google Cloud

Feedback:

A. Assign permissions to a Google account referenced by the application

Feedback: Incorrect. A Google account uses a username and password to authenticate a user. An application does not authenticate interactively with this type of account.

B. Assign permissions through a Google Workspace account referenced by the application

Feedback: Incorrect. A Google Workspace account is an account created for you as part of an organization that is using Google Workspace products to collaborate with one another. It is not appropriate for managing the permissions an application needs to communicate with a backend.

*C. Assign permissions through service account referenced by the application

Feedback: Correct! A service account uses an account identity and an access key. It is used by applications to connect to services.

D. Assign permissions through a Cloud Identity account referenced by the application

Feedback: Incorrect. Cloud Identity is a user management tool for providing login credentials to users of an organization that does not use Google Workspace collaboration tools. Cloud Identity is not used to manage application authentication.

Where to look:

<https://cloud.google.com/iam/docs/overview>

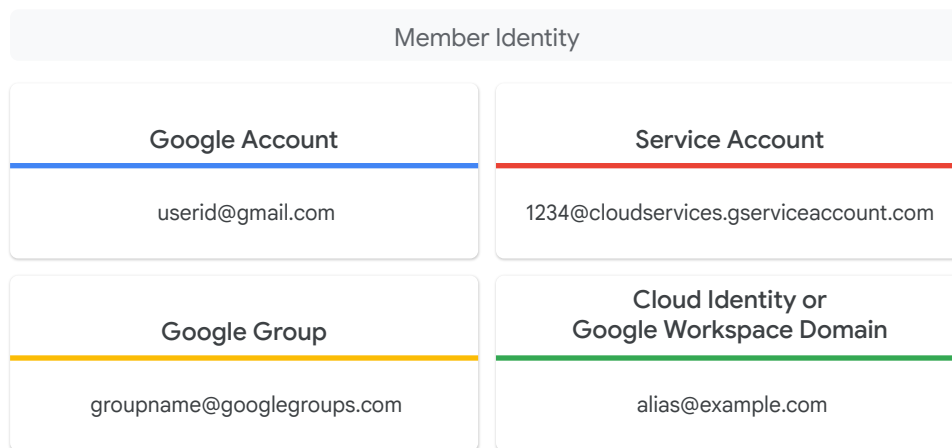
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M2 Resources and Access in the Cloud
- Architecting with Google Compute Engine (ILT)
 - M4 Identity and Access Management
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M1 Identity and Access Management
- Skill Badge: Develop your Google Cloud Network
(https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Assign access to members using IAM



A Google account represents anyone who interacts with Google Cloud. When signing up for a Google account you will be asked to provide an email address that is associated with the account. The email does not have to come from the gmail domain.

A service account is how applications and resources authenticate and access services in Google Cloud. Since apps cannot sign in interactively with a username and password, service accounts use keys to authenticate.

Google groups are collections of identity principals that can be referenced by the email address assigned to the group. You can apply access policies to a group. Each member of the group will receive the permissions you specify in the group policy as they authenticate.

Google Workspace and Cloud Identity domains give you the ability to manage users based on the way your organization interacts with Google. Each method gives you a virtual group representing all the registered users in your organization and the ability to add, modify, and delete users and groups.

5.1 Diagnostic Question 02 Discussion



You are trying to assign roles to the dev and prod projects of Cymbal Superstore's e-commerce app but are receiving an error when you try to run **set-iam policy**. The projects are organized into an ecommerce folder in the Cymbal Superstore organizational hierarchy. You want to follow best practices for the permissions you need while respecting the practice of least privilege.

What should you do?

- A. Ask your administrator for resource manager.projects.setIamPolicy roles for each project.
- B. Ask your administrator for the roles/resource manager.folderIamAdmin for the ecommerce folder.
- C. Ask your administrator for the roles/resource manager.organizationAdmin for Cymbal Superstore.
- D. Ask your administrator for the roles/iam.securityAdmin role in IAM.

Question:

You are trying to assign roles to the dev and prod projects of Cymbal Superstore's e-commerce app but are receiving an error when you try to run **set-iam policy**. The projects are organized into an ecommerce folder in the Cymbal Superstore organizational hierarchy. You want to follow best practices for the permissions you need while respecting the practice of least privilege. What should you do?

5.1 Diagnostic Question 02 Discussion



You are trying to assign roles to the dev and prod projects of Cymbal Superstore's e-commerce app but are receiving an error when you try to run **set-iam policy**. The projects are organized into an ecommerce folder in the Cymbal Superstore organizational hierarchy. You want to follow best practices for the permissions you need while respecting the practice of least privilege.

What should you do?

- A. Ask your administrator for `resourcemanager.projects.setIamPolicy` roles for each project.
- B. Ask your administrator for the `roles/resourcemanager.folderIamAdmin` for the ecommerce folder. ✓
- C. Ask your administrator for the `roles/resourcemanager.organizationAdmin` for Cymbal Superstore.
- D. Ask your administrator for the `roles/iam.securityAdmin` role in IAM.

Google Cloud

Feedback:

A. Ask your administrator for `resourcemanager.projects.setIamPolicy` roles for each project

Feedback: Incorrect. Best practice is to minimize the number of access policies you require.

*B. Ask your administrator for the `roles/resourcemanager.folderIamAdmin` for the ecommerce folder

Feedback: Correct! This choice gives you the required permissions while minimizing the number of individual resources you have to set permissions for.

C. Ask your administrator for the `roles/resourcemanager.organizationAdmin` for Cymbal Superstore

Feedback: Incorrect. This does not meet the requirements for least privilege.

D. Ask your administrator for the `roles/iam.securityAdmin` role in IAM.

Feedback: Incorrect. Security Admin allows you to access most Google Cloud resources. Assigning the security Admin role does not meet least privilege requirements.

Where to look:

https://cloud.google.com/architecture/prep-kubernetes-engine-for-prod#managing_identity_and_access

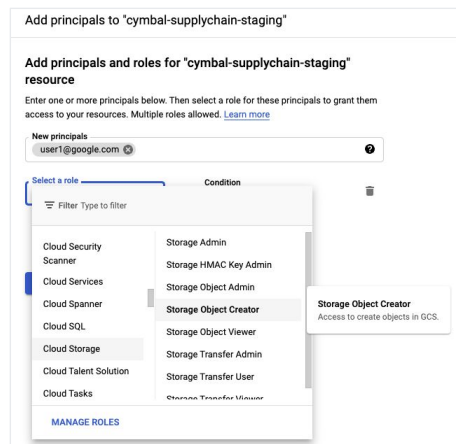
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M2 Resources and Access in the Cloud
- Architecting with Google Compute Engine (ILT)
 - M4 Identity and Access Management
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M1 Identity and Access Management
- Skill Badge: Develop your Google Cloud Network
(https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Assign roles in the IAM interface



These are the steps to assign roles in the IAM interface.

1. Go to the IAM page
2. Select project, folder, or organization
3. Show info panel if it is not available
4. Click permissions
5. Select or add a principal to add a role to
 - a. If the principal already exists click on Add another role
 - b. For a new principal click add and enter the principals email address
6. Select a role to grant
7. Add a condition
8. Click Save

5.1 | Diagnostic Question 03 Discussion



You have a custom role implemented for administration of the dev/test environment for Cymbal Superstore's transportation management application. You are developing a pilot to use Cloud Run instead of Cloud Run functions. You want to ensure your administrators have the correct access to the new resources.

What should you do?

- A. Make the change to the custom role locally and run an update on the custom role.
- B. Delete the custom role and recreate a new custom role with required permissions.
- C. Copy the existing role, add the new permissions to the copy, and delete the old role.
- D. Create a new role with needed permissions and migrate users to it.

Question:

You have a custom role implemented for administration of the dev/test environment for Cymbal Superstore's transportation management application. You are developing a pilot to use Cloud Run instead of Cloud Run functions. You want to ensure your administrators have the correct access to the new resources. What should you do?

5.1 Diagnostic Question 03 Discussion



You have a custom role implemented for administration of the dev/test environment for Cymbal Superstore's transportation management application. You are developing a pilot to use Cloud Run instead of Cloud Run functions. You want to ensure your administrators have the correct access to the new resources.

What should you do?

- A. Make the change to the custom role locally and run an update on the custom role. 
- B. Delete the custom role and recreate a new custom role with required permissions.
- C. Copy the existing role, add the new permissions to the copy, and delete the old role.
- D. Create a new role with needed permissions and migrate users to it.

Google Cloud

Feedback:

*A. Make the change to the custom role locally and run an update on the custom role

Feedback: Correct! There is a recommended process to update an existing custom role. You get the current policy, update it locally, and write the updated policy back into Google Cloud. The gcloud commands used in this process include the get and update policy subcommands.

B. Delete the custom role and recreate a new custom role with required permissions

Feedback: Incorrect. Recreating a custom role is not necessary in this scenario. You can update the existing one.

C. Copy the existing role, add the new permissions to the copy, and delete the old role

Feedback: Incorrect. Copying an existing role creates a new custom role. Creating a new custom role is not required for this scenario.

D. Create a new role with needed permissions and migrate users to it.

Feedback: Incorrect. Finding all users with this role and reassigning them could be very time consuming. You should update the existing custom role instead.

Where to look:

<https://cloud.google.com/iam/docs/creating-custom-roles>

Content mapping:

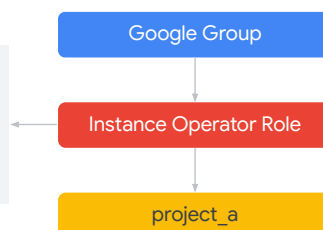
- Architecting with Google Compute Engine (ILT)
 - M4 Identity and Access Management
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M1 Identity and Access Management
- Skill Badge: Develop your Google Cloud Network (https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Create custom roles

- ✓ `compute.instances.get`
- ✓ `compute.instances.list`
- ✓ `compute.instances.start`
- ✓ `compute.instances.stop`



The first thing you need to do when creating custom permissions is be familiar with the permissions and roles that are available in your project or organization.

The `gcloud` command you need to run is:

```
gcloud iam list-testable-permissions <full-resource-name>
```

To make sure there isn't already another role that will fill your needs, you can also look at the permissions assigned to a specific role by looking at the role metadata. The role metadata includes the role ID and the permissions associated with that role.

Custom roles can be created at the project or organizational level.

You need to have the `iam.roles.create` permission. You have to be the owner of the group or project, or have an organization administrator role or the IAM Role Administrator role.

You can create roles from individual permissions, or you can select and pick permissions from predefined roles.

To update an existing role, you run `roles.get()`, update the role locally, and then run `roles.patch()`.

5.1 Managing Identity and Access Management (IAM)

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M2 Resources and Access in the Cloud

[Architecting with Google Compute Engine](#)

- M4 Identity and Access Management (IAM)

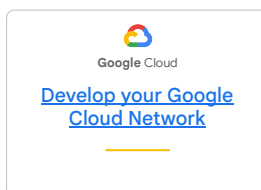


[Essential Google Cloud Infrastructure: Core Services](#)

- M1 Identity and Access Management (IAM)



Skill Badge



Documentation

[Overview | IAM Documentation](#)

[Google Kubernetes Engine security overview](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Core Services \(On-demand\)](#)

[Develop your Google Cloud Network \(Skill Badge\)](#)

<https://cloud.google.com/iam/docs/overview>

<https://cloud.google.com/kubernetes-engine/docs/concepts/security-overview>

5.2 | Managing service accounts

Considerations include:

- Creating service accounts
- Using service accounts in IAM policies with minimum permissions
- Assigning service accounts to resources
- Managing IAM of a service account
- Managing service account impersonation
- Creating and managing short-lived service account credentials

Google Cloud

Cymbal Superstore's three applications targeted for migration all have frontend and backend resources required to implement their final solutions. We've discussed what resources are required to support the backend stores, to include Spanner, Bigtable, and Cloud SQL.

There is another security concern we haven't talked about in detail. How do we enable the machine-to-machine or server-to-server access requirements for the application to talk to those backends?

Let's look at the supply chain management application for an example. The data for it is stored in Cloud SQL. The application needs permissions to write data to the Cloud SQL service.

How do you get the VMs implemented as part of the supply chain management app the permissions required to connect to the Cloud SQL database in a secure manner?

The solution to machine-to-machine access is a service account. As an Associate Cloud Engineer, you need to know how to create a service account and assign roles to it. You also need to know how to list a service account's permissions, and allow other users to inherit its permissions - also called impersonation. Finally, because they don't sign in interactively like a user does, service accounts authenticate via keys. Managing those keys and providing temporary credentials through code are important things for you to know as you help secure your cloud solutions.

These diagnostic questions addressed managing service accounts:

Question 4: Differentiate between Google accounts and service accounts in IAM.

Question 5: Identify the section of the Google API that specifies an IAM scope.

5.2 | Diagnostic Question 04 Discussion



Which of the scenarios below is an example of a situation where you should use a service account?

- A. To directly access user data
- B. For development environments
- C. For interactive analysis
- D. For individual GKE pods

Question:

Which of the scenarios below is an example of a situation where you should use a service account?

5.2 | Diagnostic Question 04 Discussion



Which of the scenarios below is an example of a situation where you should use a service account?

- A. To directly access user data
- B. For development environments
- C. For interactive analysis
- D. For individual GKE pods



Google Cloud

Feedback:

A. To directly access user data

Feedback: Incorrect. Service accounts should not be used to access user data without consent.

B. For development environments

Feedback: Incorrect. Service accounts should not be used for development environments. Use the application default credentials.

C. For interactive analysis

Feedback: Incorrect. Service accounts should be used for unattended work that does not require user interaction.

*D. For individual GKE pods

Feedback: Correct! When configuring access for GKE, you set up dedicated service accounts for each pod. You then use workload identity to map them to dedicated Kubernetes service accounts.

Where to look:

<https://cloud.google.com/docs/authentication/production#automatically>

Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)

- M2 Resources and Access in the Cloud
- Architecting with Google Compute Engine (ILT)
 - M4 Identity and Access Management
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M1 Identity and Access Management

Summary:

Explanation/summary on the following slide.

Create, use, and assign service accounts

01

To create a service account:

```
gcloud iam
service-accounts create
```

02

To assign policies:

```
gcloud projects
add-iam-policy
```

03

Attach a service account to a resource as you create it

```
gcloud compute instances create
cymbal-vm --service-account \
<name-of-service-account@gservic
eaccount.com> \
--scopes
https://www.googleapis.com/auth/
cloud-platform
```

Google Cloud

Creating a service account

https://cloud.google.com/iam/docs/creating-managing-service-accounts#creating_a_service_account

To create a service account you use the “gcloud iam service-accounts create” command.

Using service accounts with IAM policies

To add a policy to a service account run the “gcloud projects add-iam-policy-binding” command.

The “--member” argument should be a string starting with “serviceAccount:” and containing your service account id with an email address suffix of “@project_id.iam.gserviceaccount.com.” A “--role” argument contains the role you want to assign to the service account.

Assigning service accounts to resources

Resources in Google Cloud can be assigned a service account that acts as the resource’s default identity. This process is known as attaching a service account to a resource. The resource, or apps running on the resource, impersonate the attached service account to access Google Cloud APIs.

<https://cloud.google.com/compute/docs/access/create-enable-service-accounts-for-instances#using>

Multiple virtual machine instances can use the same service account, but a virtual machine can only have one service account identity. Service account changes will affect all virtual machine instances using the service account. You can allow access via a cloud-platform scope that allows access to most cloud api's and then grant the service account the relevant IAM roles.

In gcloud you identify the service account you want to use by using the "--service-account" argument.

<https://developers.google.com/identity/protocols/oauth2/service-account#python>

Two types of keys are available for authentication of a service account: user managed keys and Google managed keys. You create and manage user managed keys yourself. Google only stores the public key.

With Google managed keys Google stores both the public and private portion of the keys. Google has APIs you can use to sign requests with the private key.

5.2 Diagnostic Question 05 Discussion



Cymbal Superstore is developing a new mobile app for end users to track their delivery status in real time. The app needs to securely receive data about truck location from a Pub/Sub topic.

- A. API key
- B. OAuth 2.0 user credentials
- C. Environment provided service account
- D. Service account key

Following Google's recommended security practices for client-side applications accessing private Google Cloud data, how should the mobile app authenticate to obtain this data?

Question:

Cymbal Superstore is developing a new mobile app for end users to track their delivery status in real time. The app needs to securely receive data about truck location from a Pub/Sub topic. Following Google's recommended security practices for client-side applications accessing private Google Cloud data, how should the mobile app authenticate to obtain this data?

5.2 Diagnostic Question 05 Discussion



Cymbal Superstore is developing a new mobile app for end users to track their delivery status in real time. The app needs to securely receive data about truck location from a Pub/Sub topic.

Following Google's recommended security practices for client-side applications accessing private Google Cloud data, how should the mobile app authenticate to obtain this data?

- A. API key
- B. OAuth 2.0 user credentials
- C. Environment provided service account
- D. Service account key



Google Cloud

Feedback:

A. API key

Incorrect. API keys are for quota/billing or public APIs; they offer no identity and cannot securely access private resources.

*B. OAuth 2.0 user credentials

Correct! The app uses OAuth 2.0 to get a user token, which a secure backend validates and uses a Service Account to access Pub/Sub. This is the recommended secure pattern.

C. Environment provided service account

Incorrect. Environment-provided service accounts are only for applications running inside Google Cloud infrastructure (e.g., Cloud Run, Compute Engine), not external mobile apps.

D. Service account key

Incorrect. Embedding a Service Account private key in a client app is a critical security risk, as the key can be extracted, granting permanent access to the resource.

Where to look:

<https://cloud.google.com/docs/authentication/>

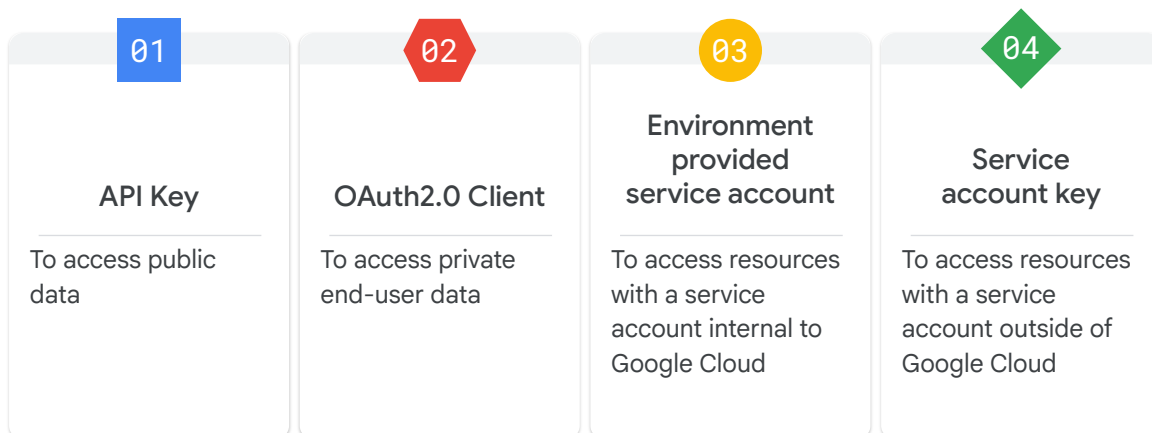
Content mapping:

- Architecting with Google Compute Engine (ILT)
 - M4 Identity and Access Management
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M1 Identity and Access Management

Summary:

Explanation/summary on the following slide.

Types of authentication keys



Google Cloud

Application credentials are determined by the type of data being accessed and the environment from which the application is running.

- If you are accessing public data, the recommendation is to use an API key. An API key identifies the application for usage and billing but does not grant access to private user information.
- If you are accessing private data on behalf of an end user, you should use the API's OAuth2.0 client flow. This delegates authority by requiring the end user's explicit consent, ensuring the application only accesses the data the user allows.
- If you are accessing private data from a resource inside Google Cloud, you should use an environment provided service account. This is the most secure and recommended method within Google Cloud as it avoids managing private keys.
- If you are accessing private data from a service account running outside of Google Cloud, you should create and use a service account key (JSON file). This is necessary when outside Google Cloud to authenticate the service account's identity. Secure storage of the key is critical.

5.2 Managing service accounts

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M2 Resources and Access in the Cloud

[Architecting with Google Compute Engine](#)

- M4 Identity and Access Management (IAM)



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[Essential Google Cloud Infrastructure: Core Services](#)

- M1 Identity and Access Management (IAM)



Documentation

[Authenticating as a service account | Authentication](#)
[Authentication overview](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Core Services \(On-demand\)](#)

<https://cloud.google.com/docs/authentication/production#automatically>

<https://cloud.google.com/docs/authentication/>